Foundations of Bias variance tradeoff

[Bias variance tradeoff]

$$\mathrm{MSE} = \mathrm{Bias}^2 + \mathrm{Variance} + \sigma_\varepsilon^2$$

1.0 Content Block

E ■ [(y − f ^ (x)) 2 ■ X = x] = (f (x) − 1 k ∑ i = 1 k f (N i (x))) 2 + σ 2 k + σ 2 {\displaystyle \operatorname {\mathbb {E} } \left[\left(y-{\hat {f}}\!(x)\right)^{2}\mid X=x\right]=\left(f(x)-{\frac {1}{k}}\sum _{i=1}^{k}f(N_{i}(x))\right)^{2}+{\frac {\sigma ^{2}}{k}}+\sigma ^{2}}

2.0 Content Block

In human learning While widely discussed in the context of machine learning, the bias–variance dilemma has been examined in the context of human cognition, most notably by Gerd Gigerenzer and co-workers in the context of learned heuristics. They have argued (see references below) that the human brain resolves the dilemma in the case of the typically sparse, poorly-characterized training-sets provided by experience by adopting high-bias/low variance heuristics. This reflects the fact that a zero-bias approach has poor generalizability to new situations, and also unreasonably presumes precise knowledge of the true state of the world. The resulting heuristics are relatively simple, but produce better inferences in a wider variety of situations. Geman et al. argue that the bias–variance dilemma implies that abilities such as generic object recognition cannot be learned from scratch, but require a certain degree of "hard wiring" that is later tuned by experience. This is because model-free approaches to inference require impractically large training sets if they are to avoid high variance.

3.0 Content Block

The expectation ranges over different choices of the training set D = { (x 1 , y 1) … , (x n , y n) } {\displaystyle D=\{(x_{1},y_{1})\dots ,(x_{n},y_{n})\}} , all sampled from the same joint distribution P (x , y) {\displaystyle P(x,y)} which can for example be done via bootstrapping. The three terms represent:

4.0 Content Block

E D , ε [(y − f ^ (x ; D)) 2] = (Bias D ■ [f ^ (x ; D)]) 2 + Var D ■ [f ^ (x ; D)] + σ 2 {\displaystyle \mathbb {E} _{D,\varepsilon }{\Big [}{\big (}y-{\hat {f}}\!(x;D){\big)}^{2}{\Big]}={\Big (}\operatorname {Bias} _{D}{\big [}{\hat {f}}\!(x;D){\big]}{\Big)}^{2}+\operatorname {Var} _{D}{\big [}{\hat {f}}\!(x;D){\big]}+\sigma ^{2}}

5.0 Content Block

Derivation The derivation of the bias–variance decomposition for squared error proceeds as follows. For convenience, we drop the D {\displaystyle D} subscript in the following lines, such that f ^ (x ; D) = f ^ (x) {\displaystyle {\hat {f}}\!(x;D)={\hat {f}}\!(x)} . Let us write the mean-squared error of our model:

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k-nearest neighbors In the case of k-nearest neighbors regression, when the expectation is taken over the possible labeling of a fixed training set, a closed-form expression exists that relates the bias–variance decomposition to the parameter k:

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MSE (x) ■ E [(y − f ^ (x)) 2] = E [(f (x) + ε − f ^ (x)) 2] since y ■ f (x) + ε = E [(f (x) − f ^ (x)) 2] + 2 E [(f (x) − f ^ (x)) ε] + E [ε 2] {\displaystyle {\begin{aligned}{\text{MSE}}(x)&\triangleq \mathbb {E} {\Big [}{\big (}y-{\hat {f}}\!(x){\big)}^{2}{\Big]}\\&=\mathbb {E} {\Big [}{\big (}f(x)+\varepsilon -{\hat {f}}\!(x){\big)}^{2}{\Big]}&&{\text{since }}y\triangleq f(x)+\varepsilon \\&=\mathbb {E} {\Big [}{\big (}f(x)-{\hat {f}}\!(x){\big)}^{2}{\Big]}\,+\,2\ \mathbb {E} {\Big [}{\big (}f(x)-{\hat {f}}\!(x){\big)}\varepsilon {\Big]}\,+\,\mathbb {E} [\varepsilon ^{2}]\end{aligned}}}

8.0 Content Block

where N 1 (x) , … , N k (x) {\displaystyle N_{1}(x),\dots ,N_{k}(x)} are the k nearest neighbors of x in the training set. The bias (first term) is a monotone rising function of k, while the variance (second term) drops off as k is increased. In fact, under "reasonable assumptions" the bias of the first-nearest neighbor (1-NN) estimator vanishes entirely as the size of the training set approaches infinity.

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E [(f (x) − f ^ (x)) ε] = E [f (x) − f ^ (x)] E [ε] since ε is independent from x = 0 since E [ε] = 0 {\displaystyle {\begin{aligned}\mathbb {E} {\Big [}{\big (}f(x)-{\hat {f}}\!(x){\big)}\varepsilon {\Big]}&=\mathbb {E} {\big [}f(x)-{\hat {f}}\!(x){\big]}\ \mathbb {E} {\big [}\varepsilon {\big]}&&{\text{since }}\varepsilon {\text{ is independent from }}x\\&=0&&{\text{since }}\mathbb {E} {\big [}\varepsilon {\big]}=0\end{aligned}}}

10.0 Content Block

In regression The bias–variance decomposition forms the conceptual basis for regression regularization methods such as LASSO and ridge regression. Regularization methods introduce bias into the regression solution that can reduce variance considerably relative to the ordinary least squares (OLS) solution. Although the OLS solution provides non-biased regression estimates, the lower variance solutions produced by regularization techniques provide superior MSE performance.

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E [(f (x) − E [f ^ (x)]) 2] = E [f (x) 2] − 2 E [f (x) E [f ^ (x)]] + E [E [f ^ (x)] 2] = f (x) 2 − 2 f (x) E [f ^ (x)] + E [f ^ (x)] 2 = (f (x) − E [f ^ (x)]) 2 {\displaystyle {\begin{aligned}{\color {Blue}\mathbb {E} {\Big [}{\big (}f(x)-\mathbb {E} {\big [}{\hat {f}}(x){\big]}{\big)}^{2}{\Big]}}&=\mathbb {E} {\big [}f(x)^{2}{\big]}\,-\,2\ \mathbb {E} {\Big [}f(x)\ \mathbb {E} {\big [}{\hat {f}}(x){\big]}{\Big]}\,+\,\mathbb {E} {\Big [}\mathbb {E} {\big [}{\hat {f}}(x){\big]}^{2}{\Big]}\\&=f(x)^{2}\,-\,2\ f(x)\ \mathbb {E} {\big [}{\hat {f}}(x){\big]}\,+\,\mathbb {E} {\big [}{\hat {f}}(x){\big]}^{2}\\&={\Big (}f(x)-\mathbb {E} {\big [}{\hat {f}}(x){\big]}{\Big)}^{2}\end{aligned}}}

12.0 Content Block

In reinforcement learning Even though the bias–variance decomposition does not directly apply in reinforcement learning, a similar tradeoff can also characterize generalization. When an agent has limited information on its environment, the suboptimality of an RL algorithm can be decomposed into the sum of two terms: a term related to an asymptotic bias and a term due to overfitting. The asymptotic bias is directly related to the learning algorithm (independently of the quantity of data) while the overfitting term comes from the fact that the amount of data is limited.